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15.1 Conclusion of EIA Study

15.1.1.1 This EIA Report has provided an assessment of the potential environmental impacts associated with the construction and operation of the Project, based on the engineering design information available during the course of EIA study.

15.1.1.2 The assessment has been conducted, in accordance with the Study Brief No. ESB-373/2025 under the EIAO and the EIAO-TM, covering the following environmental issues:

- Air Quality Impact;
- Noise Impact;
- Water Quality Impact;
- Waste Management Implications;
- Land Contamination;
- Ecological Impact (Terrestrial and Aquatic);
- Fisheries Impact;
- Landscape and Visual Impact;
- Cultural Heritage; and
- Hazard to Life.

15.1.1.3 The findings of this EIA study have determined the likely nature and extent of environmental impacts predicted to arise from the construction and operation of the Project. During the EIA process, environmental control measures have been identified and incorporated into the planning and design of the Project, to achieve compliance with environmental legislation and standards during both the construction and operational phases. The Implementation Schedule listing the recommended mitigation measures is presented in the stand-alone Environmental Monitoring & Audit Manual.

15.1.1.4 The various sections of this EIA Report have presented the measures to minimise the potential environmental impacts associated with the Project throughout the construction and operational phases. The key measures to minimise the environmental impacts are summarized in **Section 14**.

15.1.1.5 With the implementation of the mitigation measures, the Project would be environmentally acceptable with no unacceptable residual impacts on the environmentally sensitive receivers and resources. **Table 15.1** below summarises the key environmental outcomes and benefits that have accrued from the environmental considerations and analysis during the EIA process and the implementation of environmental control measures of the Project. **Appendix 15.1** presents the summary of key assessment assumptions, limitation of assessment methodologies, while **Appendix 15.2** presents the summary of environmental impacts.

15.1.1.6 An Environmental Monitoring and Audit (EM&A) programme has been developed to ascertain and verify the assumptions implicit to, and accuracy of, EIA study predictions. EM&A requirements have been recommended, where necessary, to check on project compliance of environmental legislation and standards. These are presented in a separate stand-alone EM&A Manual.

Table 15.1 Summary of Key Environmental Outcomes and Benefits

Environmental Issue	Description
Environmental / Social Benefits of the Project and Environmentally Friendly design	• Avoidance of direct impact to ecologically sensitive areas - The recommended alignment and its works footprint have been thoroughly considered and evaluated from alternative alignment options proposed in previous stage studies (**Section 2.6 of Chapter 2** refers). This process has resulted in the recommended alignment option, which aims to avoid potential both direct and indirect impacts on recognised sites of conservation importance and other ecologically sensitive areas, such as the Lam Tsuen Country Park (LTCP), the proposed Nam Sang Wai Wetland Conservation Park and natural watercourses at Ngau Tam Mei. The locations of piers are also carefully designed to avoid direct encroachment into natural watercourses. • Wetland Habitat Enhancement through Integration and Expansion – Under the original alignment, an interchange, I/C-03, (with associated site office and ancillary facilities) was proposed to the west of Ngau Tam Mei Water Treatment Works (NTMWTW), where scattered and fragmented wetland habitats are currently found. Compared to the original alignment, the recommended alignment has been refined, with the Interchange I/C-03 relocated to the south of Ngau Tam Mei New Development Area (NTM NDA). The viaduct level has also been raised as much as practicable to provide greater clearance allowing natural sunlight to reach the habitats underneath the viaduct structures. The relocation of Interchange I/C-03 to the south of NTM NDA will also provide an opportunity to integrate and expand the wetland habitats to the west of NTMWTW. Furthermore, the mainline will be constructed as an elevated viaduct with a limited number of piers, spanning over the wetland habitats to reduce direct ecological impact to this wetland habitats. Taking advantage of the Project implementation, the area will not only be reinstated, but the existing fragmented wetland habitats will also be integrated and enhanced into a connected wetland habitat. The total wetland area will increase from approximately 3 hectares of fragmented ponds to more than 5 hectares of integrated wetland habitats through the incorporation of adjacent village / orchard land. With improved design and management, the enhanced wetland habitat will support higher biodiversity, greater ecological value and increased carrying capacity. It will also offer a suitable foraging environment for various relevant species, including waterbirds. • Allowing Wildlife Movement by Adopting Intermittent At-grade and Viaduct Design – A section of the alignment runs adjacent to the woodland habitat at the edge of the Lam Tsuen Country Park (LTCP) near Long Ha. The alignment profile has been raised as far as practicable, taking into account site constraints such as CLP cables, pylons and traffic design requirements. The design has been optimised to avoid encroachment into LTCP while

Environmental Issue	Description
	maximising the length of the viaduct section. Each of the at-grade sections is no greater than ~350 m so as to avoid creating potential hurdles to wildlife movement. The space beneath the viaduct sections provides an open corridor to facilitate wildlife movement to and from LTCP. For the at-grade sections, retaining wall on the slope would be constructed to avoid wildlife roadkill. • Viaduct Level Arrangement and Carriageway Separation for Enhancing Sunlight Penetration and Ecological Benefits – The Project includes slip roads crossing the Kam Tin River, where the viaduct level has been adjusted to provide headroom of about 10~15m and minimise shading effects on the water body and associated habitats. In the vicinity of NTMWTW, to accommodate the required clearance from the existing CLP cable, the headroom beneath the viaduct is approximately 10m for the northbound and ranges from 5m to 10m for the southbound. The dual carriageways are separated before entering the tunnel portal. The resulting gap between the carriageways enables natural sunlight to reach the ground below and is not expected to cause adverse ecological impacts. Furthermore, the positioning of piers have been refined to avoid encroachment into natural watercourses. This approach helps to minimize potential water quality and ecological impact. • Adoption of non-dredged method for construction works at Kam Tin River - The significance of ecological value of Kam Tin River is well recognised. Given the large span of the viaduct crossing Kam Tin River, the pile /pier support within the river is inevitable. The construction of viaduct pier/ pile will adopt the non-dredging method. For example, if bored pile construction is adopted, steel casing could be adopted to isolate the surrounding riverbed. The material within the steel casing will be excavated within the casing for placing reinforcement and concrete of the pile. This minimises the disturbance to the river. • Synergistic Woodland Enhancement near She Leng – A woodland compensation site of approximately 17 ha is proposed near She Leng, which is currently characterised by grassland or shrubland of low or low to moderate ecological value. The site is also adjacent to proposed woodland compensation areas under other projects, including Lok Ma Chau Loop, San Tin Technopole (STT) and Development of the Loop – Eastern Connection Road (ECR). Woodland compensation at this location will integrate the overall compensatory woodland area and improve habitat quality and ecological connectivity with adjacent woodland habitats. • Minimisation of slope cutting through adoption of elevated alignment – Given the hilly terrain to the south of NTM NDA, the road alignment has been elevated. This adjustment not only minimises the scale of site formation works, but also takes into account other constraints, such as the clearance requirement for CLP power cables. Additionally, a larger headroom would provide greater clearance between the structure and the habitats so as to

Environmental Issue	Description
	allow natural sunlight to shine on the habitats underneath the viaduct structures. • Minimisation of slope cutting through soft ground tunnelling at portals – When the alignment passes through Ngau Tam Shan and Hadden Hill, soft ground tunnelling has been adopted at portals. Specifically, before the tunnel excavation reaches bedrock, soft soil stratum will be improved with injection grouting around the tunnel profile. Temporary support such as pipe roofing and steel lattice girder/rib will be installed during the tunnel excavation by mechanical method. With the adoption of soft ground tunnelling, the extent of site formation is significantly reduced, which helps retain a larger area of natural terrain and minimise the generation of construction and demolition (C&D) materials. • Minimisation of slope cutting through construction of retaining walls and soil nailing – For sections where extensive cut slopes would otherwise be required, retaining walls / soil nails or combination of both may be proposed. This measure aims to reduce the extent of slope excavation, allow for steeper slope gradients, and thereby minimise the project's environmental footprint, as well as acting as a barrier to prevent wildlife roadkill. • Avoidance of direct impacts on Sites of Archaeological Interest (SAI) – Avoid any excavation works which will cause potential disturbance to the soil matrix within the boundary of Ngau Tam Mei SAI to avoid potential direct impacts on the SAI. • Adopt Drill and Blast (D&B) method for tunnel construction – Adopt D&B method over TBM (Tunnel Boring Machine) method for tunnel construction to minimize the generation of construction & demolition (C&D) materials as much as practicable. • Diverting the existing traffic flow between northeast and northwest of the New Territories, lowering vehicular emissions from idling or congested traffic and shortening travel times and results in further reducing fuel consumption, alleviating potential air quality and noise impact during operation phase.
Air Quality Impact	<u>Construction Phase</u> • Potential air quality impact from the construction works of the Project would mainly be related to construction dust from site formation, site clearance, excavation, slope works, construction of administration buildings and satellite control buildings, blasting for tunnel construction, mucking out blasted rock and soil from drill-and-blast operations, material handling and backfilling, foundation works, handling and stockpiling of dusty materials, wind erosion of exposed area, as well as use of on-site powered mechanical equipment, and emissions from construction vehicles. Construction activities of the concurrent projects within 500m assessment area would also pose cumulative air quality impact. With the implementation of mitigation measures specified in the *Air Pollution Control (Construction Dust)*

Environmental Issue	Description
	Regulation together with the recommended control measures including frequent watering on active works areas, exposed areas and unpaved haul roads and other site management measures such as good site practices, use of electric vehicles and EM&A programme, no adverse air quality impact on ASRs would be anticipated during the construction stage. <u>Operation Phase</u> • Cumulative air quality impacts were assessed by considering the Project-induced emissions, all relevant emission sources within the 500 m assessment area, such as vehicular emission from open roads, tunnel portals and facilities with vehicular emissions (existing and planned public transport interchanges, bus and minibus Termini, heavy goods vehicle and coach parking sites), industrial emissions, and background concentrations. The quantitative modelling prediction results concluded that the cumulative NO_2, RSP, and FSP concentrations at all existing and planned ASRs would comply with the prevailing AQOs. Thus, no adverse air quality impact on the existing and planned ASRs would be anticipated.
Noise Impact	<u>Construction Noise</u> • Assessment on potential construction noise impact arising from the Project has been conducted qualitatively. The assessment results indicate that with the implementation of appropriate mitigation measures, no adverse construction noise impact would be anticipated. <u>Road Traffic Noise</u> • While for the road traffic noise, the assessment results indicate that the predicted road traffic noise levels at some of the representative Noise Sensitive Receiver (NSRs) would exceed the noise criteria under the unmitigated scenario. Direct noise mitigation measures have been considered to alleviate the potential traffic noise impact. With implementation of the recommended noise mitigation measures of Low Noise Road Surfacing and reflective and absorptive vertical and cantilevered type noise barrier and the provision of at-receiver mitigation measures, such as acoustic windows/balconies, the noise levels at all NSRs within the assessment area would comply with the traffic noise criteria stipulated in Clause 4.7.6 of EIAO GN No. 12/2023. As such, adverse road traffic noise impact due to the Project is not anticipated. <u>Fixed Noise Sources</u> • In view of the large separation distance between the

Environmental Issue	Description
	planned fixed noise sources and the noise sensitive receivers, and that the impact from the planned fixed noise sources could be effectively mitigated by implementing at-source noise control measures during the detailed design stage, no adverse impact from both the proposed fixed noise sources of the Project would be anticipated.
Water Quality Impact	• Construction of pier and viaduct structures across Kam Tin River would require installation of cofferdam and/or watertight pile casing to enclose the associated construction works. Cofferdam and watertight pile casing should be constructed to isolate the construction activities from the river water. Cofferdam and watertight pile casing should be designed by the Contractor which fulfil the requirements in DSD Technical Circular No. 1/2017 "*Temporary Flow Diversions and Temporary Works Affecting Capacity in Stormwater System*". Water pumps should be used to collect any construction site runoff and ingress/seepage water within the cofferdam and watertight pile casing to the on-site wastewater treatment facilities for treatment to satisfactory levels before discharge. Silt curtain should be deployed to completely enclose the cofferdam and watertight pile casing prior to conduct construction works in Kam Tin River, and installation and removal of cofferdam and/or watertight pile casing. • Pier construction at Kam Tin River should be conducted by phases to avoid adverse water flow impact due to flow blockage increase during the interim construction stages. • Water quality impacts from land-based construction would be controlled by implementing the recommended mitigation measures and good site practices. Water quality monitoring and regular site inspections would be carried out routinely to minimise water quality impacts upon the nearby inland waters. An Emergency Response Plan is recommended to minimise the potential water quality impact from construction site discharges under failure of treatment facilities during emergency situations or inclement weather. • Potential water quality impacts during operation phase were related to the non-point source surface runoff generated from the additional paved areas, change in flow regime due to the presence of bridge piers in Kam Tin River, as well as sewage effluent generated from the new administration building. Provided that the recommended mitigation measures for the drainage system are properly implemented and the sewage generated from the new administration building to be properly collected, treated and disposed, the associated water quality impacts were expected to be minimal and acceptable. No significant change in flow regime and water quality in Kam Tin River was expected and therefore, no adverse water flow and water quality impacts on Kam Tin River was expected.

Environmental Issue	Description
Waste Management Implications	Construction Phase • C&D materials, excavated sediments, chemical waste and general refuse would be generated during the construction phase. Provided that these wastes are handled, transported and disposed of according to the recommended good site practices and measures, no adverse environmental impacts (including potential hazard, air and odour emissions, noise and wastewater discharges) would be anticipated during the construction phase. Operation Phase • General refuse and chemical waste would be generated during the operation phase. With the implementation of management control practices, no adverse environmental impact would be anticipated.
Land Contamination	• Site appraisals, in the form of desktop review and site walkovers, had been carried out between September 2025 and January 2026 to identify the past and current potentially contaminating land uses within the Project Site. Based on the site appraisal, 30 potentially contaminated sites, 15 sites suspected to be used for industrial purposes were identified within the Project Site. • Further land contamination assessments, including further site appraisal and submission of CAP(s), should be conducted for the whole Project Site at a later stage of the Project when site access is available to confirm the existing land uses / activities, identify the presence of any potential contamination sources, and address any new contamination issues. The associated SI works and any necessary remediation action are recommended to be carried out after operation of concerned site(s) has ceased but prior to the commencement of construction works. The recommended further assessment and remediation works, including the submission of CAP(s), CAR(s)/RAP(s) and RR(s) would follow the relevant Guidance Manual, Guidance Note and Practice Guide. • With reference to previous studies, natural occurrences of elevated level of arsenic exist over broad areas of the northern New Territories. A separate arsenic assessment and remediation plan should be prepared, in consultation with EPD, and submitted to EPD for endorsement prior to the commencement of construction works. • With the implementation of the recommended further works for the Project, any soil/groundwater contamination would be identified and properly remediated prior to the construction works. No insurmountable land contamination impacts to the Project are therefore anticipated.
Ecology	• The potential ecological impacts arising from the construction works of the Project would include habitat loss, disturbance to terrestrial habitats and associated wildlife, potential obstruction of flight paths, mortality risks to species of conservation importance and other indirect impacts on sites of conservation importance, natural habitats and associated wildlife in the vicinity. With the

Environmental Issue	Description
	implementation of avoidance, minimisation and compensation measures recommended in Ecological Impact Assessment, no adverse ecological impact is anticipated.
Fisheries	• The occupation or suspension of fisheries operation is of negligible significance to local aquaculture production. With the implementation of good site practices specified in the EIA, no adverse fisheries impacts is anticipated.
Landscape and Visual	• Among the approx. 67,500 nos. of existing trees within the 100m assessment area, it is estimated that there are 9,050 nos. of trees (excluding all *Leucaena leucocephala* of which removal would not require a TPRP) would be affected. • Under this Project, the existing landscape resources, including vegetation, marsh or wetland and fishpond would be affected in multiple scale subject to the proposed works. Vegetation found within the LRs and LCAs would either be temporarily or permanently loss. After the completion of works, landscape mitigation measures such as tree compensation where space is available, reinstatement of affected landscaping area or fishpond in like for like basis and provision of buffer screen planting would be implemented. With the implementation of these mitigation measures, it is considered that residual impacts of the LRs and LCAs would be reduced to Negligible to Slight in the operational phase. • In terms of the visual impact, considering the connecting alignment to the existing infrastructures and structural mass of proposed road works, the visual impact to the identified key public viewers would be ranging from Negligible to Moderate. • Considering the scale and nature of the Project, it would inevitably result in certain levels of residual landscape impacts and visual impacts in relation to the loss of hillside vegetation from hilltop and from ground level. Nevertheless, the residual landscape impacts are localized and limited to the project extent only without affecting existing community, while the visual impacts are confined within the visual envelope either involving few to medium numbers of public viewers along hiking trail and paths. With the implementation of the proposed landscape and visual mitigation measures, the overall landscape residual impacts would be from Negligible to Moderate during construction phase, and Negligible to Slight in operational phase, and the overall visual impacts would be from Negligible to Moderate in operational phase. With full implementation of the recommended mitigation measures, unacceptable adverse residual landscape impacts and visual impacts are not expected.
Cultural Heritage	• No declared monument, proposed monument, graded historic building, new item for grading assessment and Government historic site identified by the Antiquities and Monuments Office (AMO) is located within the Project

Environmental Issue	Description
	Boundary. Therefore, no direct impact is anticipated on them during the construction and operational phases. • There are five (5) grade 2 historic buildings and eight (8) grade 3 historic buildings situated at a considerable distance from the Project Boundary within the assessment area. Therefore, no direct or indirect impact is anticipated on these historic buildings during the construction and operational phases. • Ngau Tam Mei SAI is partially located within the Project Boundary but outside the proposed alignment. According to the latest design of the construction works arrangement, no sub-soil disturbance works will be carried out within the SAI. Therefore, no impact to the SAI is anticipated. • If the work arrangement deviates from the current proposed situations, an archaeological survey-cum-excavation should be carried out to obtain archaeological data for assessing the impact before commencement of the works, in consultation with the AMO, and to design mitigation measure as appropriate. • Pursuant to the A&MO (Cap. 53), the project proponent is required to inform AMO immediately in case of discovery of antiquities or supposed antiquities in the course of works, so that appropriate mitigation measures, if needed, can be timely formulated and implemented in agreement with and to the satisfaction of AMO.
Hazard to Life	• The assessment results show that the individual risks associated with the storage, transport and use of explosives complied with the Risk Guidelines as stated in Annex 4 of the EIAO-TM. • The societal risks associated with the storage and use of explosives fall into the “Acceptable” region of the Risk Guidelines; while the societal risk associated with the transport of explosives falls into the lower “ALARP” region of the Risk Guidelines, mitigation measures for the transport of explosives were explored and assessed by using CBA. Justified mitigation measures and good practices were recommended.